APSCO

LINKU Project

Camera System Working Document

"Integrated Instrument, Dark Studio, Tether in Motion"

October 21, 2026

The LINKU TV2 Experiment: Protocol, Parameters and Record

Prepared by:

[1] Pontificia Universidad Catolica del Peru (PUCP)

Author(s): Jose Sifuentes[1]

Status: Protocol and acceptance criteria registered, session not yet recorded

Issue: 1
Revision: 0

Abstract

This document is the operating record of the LINKU TV2 experiment: the first end-to-end run of the camera payload on the integrated support, inside the blacked-out television studio of the PUCP Faculty of Communications, imaging a physical tether in motion. It states what the experiment must establish, fixes every parameter that has to be set before the room is entered, gives the order in which the session is run, and defines in advance what will count as a pass.

The document exists because three of the project's other deliverables each defer the same decision to a future self: none of them states a quantitative threshold against which a result would be judged, and the camera system report carries no reconstruction accuracy figure at all. Chapter 2 closes that gap for this experiment by registering the criteria *before* the data exists, which is the only point at which they can be set honestly.

Chapter 1 states the objectives and what each one is evidence for. Chapter 2 registers the acceptance criteria. Chapter 3 describes the facility and the instrument as built. Chapter 4 collects every parameter to be set and locked, each with its derivation and its source. Chapter 5 gives the session procedure, whose ordering is load-bearing. Chapter 6 defines what is written down and how the captures are named, so the session can be replicated in Blender afterwards. Chapter 7 specifies the analysis to be run on the captures.

Status. Everything that does not depend on results is complete. The results chapter is empty by construction: at the time of writing no capture from this experiment exists in the repository, and the record sheet of the pre-session parameter artifact is blank. Chapter 8 is the place those go, and it states exactly what must be filled in.

Contents

Introduction 6

1 Objectives 8
- 1.1 O1. Establish a valid calibration on the integrated support . . . 8
- 1.2 O2. Confirm the illumination reproduces the orbital condition . . . 8
- 1.3 O3. Run the full pipeline end to end on a real moving tether . . . 8
- 1.4 O4. Record the scene well enough to rebuild it in Blender . . . 9
- 1.5 What this experiment deliberately does not do . . . 9

2 Acceptance criteria, registered in advance 10
- 2.1 Why this chapter is not optional . . . 10
- 2.2 A1. Calibration . . . 10
- 2.3 A2. Illumination . . . 11
- 2.4 A3. Pipeline . . . 12
- 2.5 A4. Record . . . 12
- 2.6 Overall verdict . . . 13

3 Facility and instrument 14
- 3.1 The venue . . . 14
- 3.2 The instrument, as built . . . 14
- 3.3 The subject . . . 15
- 3.4 Working distance and what the optics do at it . . . 15

4 Parameters to set and lock 16
- 4.1 Illumination . . . 16
- 4.2 The penumbra test . . . 17
- 4.3 Optics . . . 17
- 4.4 Calibration . . . 18
- 4.5 Camera settings during the checks and the run . . . 18

5 Session procedure 19
- 5.1 Why the order matters . . . 19
- 5.2 The ten steps . . . 19
- 5.3 Running the calibration . . . 20
- 5.4 The capture plan . . . 20

6 What gets recorded 22
- 6.1 The record sheet . . . 22
- 6.2 Why the scene geometry fields are there . . . 23
- 6.3 Files, naming and metadata . . . 23
- 6.4 What is deliberately not recorded . . . 24

7 Analysis plan 25
- 7.1 The pipeline, stage by stage . . . 25
- 7.2 What is computed, and against what . . . 25
- 7.3 The retained fraction has to be reported with every length . . . 25
- 7.4 The comparison that is expected to fail, and why that is useful . . . 26

7.5 Why dense stereo is not used, and the trap that creates . . . 26
7.6 Feeding the datasets . . . 26

8 Results **28**
8.1 Session record . . . 28
8.2 Calibration outcome . . . 28
8.3 Illumination outcome . . . 28
8.4 Pipeline outcome . . . 28
8.5 The two anticipated failure modes . . . 28
8.6 Verdict . . . 29
8.7 What the next session should change . . . 29

Status and limitations **30**

A Constants and their provenance **32**

B Sources, with their provenance level **34**

List of Figures

5.1 Session procedure 19

List of Tables

2.1 Calibration acceptance criteria 10
2.2 Illumination acceptance criteria 11
2.3 Pipeline acceptance criteria 12
2.4 Record acceptance criteria 13

3.1 Instrument configuration 14
3.2 Optical behaviour at the working distance 15

5.1 Capture plan 21

6.1 Session record sheet 22

7.1 Analysis outputs 26

8.1 Status of this document 30

A.1 Constants used 32

B.1 Source provenance 34

Introduction

The LINKU camera payload measures a thin tether strung between two satellites. It returns metric scalars about that tether, its arc length, straightness, depth range, diameter and curvature, and a sparse point cloud as a by-product. Every stage of the processing chain has been built and exercised, and the optical and photometric requirements have been derived from primary sources in the companion deliverables. What has not happened is the thing all of it exists for: the instrument has never been pointed at a real tether, in the dark, on the support it will actually fly on, with the tether and the rig both in motion.

That is this experiment. In the project work plan [1] it is the package *Experimento de caja negra, hilo en movimiento*, inside the phase-2 deliverable *Integracion y experimento en la estructura LINKU*. It is run in a blacked-out television studio of the PUCP Faculty of Communications, referred to throughout this document as **TV2**, a space borrowed for the purpose because it offers what the project needs and does not own: a room that can be made genuinely dark, with professional lighting fixtures and the rigging to position them.

Why this document exists

Three reasons, in order of importance.

First, the acceptance criteria have to be written down before the data exists. This is not a point of style. A review of the project's own deliverables shows the same gap in three of them independently: the rolling-shutter experiment and the Blender validation each carry a pending item instructing a future author to define the quantitative agreement threshold before running, rather than after seeing the results; the calibration method leaves three of its four acceptance criteria numerically unset; and the camera system report [2] contains no reconstruction accuracy figure at all, reporting its two reconstruction scenarios qualitatively. The common root is that no mission accuracy budget for tether reconstruction has been written anywhere in the project. This document cannot supply that budget, but it can refuse to inherit the habit. Chapter 2 registers the criteria for this experiment in advance.

Second, the parameters are scattered and some of them are destructive to get wrong. The illumination requirements live in one deliverable, the optical discipline in another, the calibration preconditions in a third, and the angular-rate limits in a fourth. On the day, in a borrowed room, with a time slot, nobody is going to open four documents. Chapter 4 collects all of them in one place, each with its derivation and its source, and marks which are physics and which are project decisions open to revision.

Third, the session has to be replicable in Blender afterwards. The Blender validation deliverable [3] compares a real photograph against a render of the same scene, and it currently cannot proceed because the laboratory-condition light values in its scene are placeholders and the physical pose of the tether in the chosen reference pair was never written down. Chapter 6 defines the record so that this experiment does not repeat that omission.

What this document is not

It is not a specification of the illumination; that is derived from primary sources in the dark-room deliverable [4] and is only summarised here. It is not a calibration procedure; that is specified and

implemented in the calibration deliverable [5] and is invoked here as a precondition. It is not an analysis of rotation effects with a known angular rate; that requires a motorised stage and is a separate experiment [6]. In TV2 both the tether and the rig are moved *by hand*, erratically and without instrumentation of the motion, which is a deliberate choice discussed in Section 1.5.

Status of this revision

Chapters 1 through 7 are complete and do not depend on results. Chapter 8 is empty: at the time of writing the repository contains no capture from this session, the most recent non-simulated stereo pairs dating from February 2026, and the record sheet of the pre-session parameter artifact has not been filled in. Chapter 8 states precisely what goes in it.

Chapter 1

Objectives

The experiment has four objectives. They are listed in the order in which a failure of one invalidates the ones below it, which is also the order in which the session is run.

1.1 O1. Establish a valid calibration on the integrated support

The stereo geometry that every reconstructed number depends on has to be measured on the support that will carry the cameras during the experiment, not on any other mount. The archived calibration is unusable and the project knows why: measured over 424 archived frames the median board tilt is 4.4°, only 39% of frames exceed 5°, and not one frame in the entire archive reaches 45°, which leaves the focal length mathematically unrecoverable [5]. The guided tool built in response refuses to report a session complete until the accepted views cover four tilt bins.

This objective is a precondition, not a result. If it fails, objectives O2 to O4 still produce images, but every metric derived from them is scaled by an unknown factor and no distance in metres can be quoted.

1.2 O2. Confirm the illumination reproduces the orbital condition

One direction of light, no fill, no bounce, a parallel enough beam, and enough of it. The requirements and their derivation are in the dark-room deliverable [4]; the measurement of whether the TV2 fixtures actually meet them has never been made, because the room has never been occupied. The enclosure used until now contains one unspecified bulb whose irradiance, spectrum and beam behaviour have never been measured.

1.3 O3. Run the full pipeline end to end on a real moving tether

Segment, find the endpoints, trace the path in both images, rectify, match, triangulate, and compute the metric scalars. Every stage has been exercised on archived or simulated data. None has been exercised on a tether that was moving, in a dark room, with the rig also moving.

Two specific failure modes are being looked for, both of which the project has predicted but never observed in hardware:

- **The parametric matching assumption.** Stage 5 pairs the point at $i\%$ along the left path with the point at $i\%$ along the right path. This holds only if both traced paths start and end at the same physical points on the tether and traverse it at the same rate. When an endpoint is occluded, or a path leaves the frame, the error is systematic and silent: there is

no epipolar check to catch it. A hand-moved tether is the first situation in which an endpoint can plausibly leave the frame mid-sequence.

- **Rolling-shutter shear.** At the software's current 15 frame/s configuration the 3-pixel shear limit corresponds to roughly 1.05°/s [6]. Hand motion will exceed that. The experiment does not measure the rate, so it cannot quantify the shear; what it can establish is whether the tracker and the matching survive it at all, and at what point they visibly stop doing so.

1.4 O4. Record the scene well enough to rebuild it in Blender

The Blender validation deliverable compares a real photograph against a render of the same scene, and it is currently blocked on two things this session can supply: measured laboratory-condition light values, which stand in the scene as placeholders, and the physical pose of the tether, which for the existing reference pair was never written down [3]. Chapter 6 is written to close both.

1.5 What this experiment deliberately does not do

It does not measure angular rate. There is no servo, no motor and no rotation stage. Both the tether and the camera rig are moved by hand, erratically. A quantitative rolling-shutter characterisation needs a commanded or encoded rate, and that is a separate experiment scheduled in a later phase [6]. Running it here with uninstrumented motion would produce numbers that look quantitative and are not, which is worse than producing none.

The choice of hand motion is not merely a concession to the absence of a rig. The deployment it stands in for is itself not a smooth commanded rotation, and the question this experiment asks is the coarser one that has to be answered first: does the pipeline survive uncontrolled motion, or does it fail, and where.

It does not attempt a reconstruction accuracy figure against ground truth. No independently measured tether geometry is available at the scale and in the pose used. What can be checked is internal consistency and physical plausibility, which is what Chapter 2 registers.

It does not use the flight sensor. The IMX708 B0308 is the decided flight sensor but has not been purchased, with procurement expected the following year. This experiment uses the IMX477 pair that exists. Every optical figure quoted in this document is therefore computed for the IMX477 with its 1.55 µm pixel and the fitted 8 mm lens, and does not transfer to the B0308.

Chapter 2

Acceptance criteria, registered in advance

This chapter is written before the session. Its thresholds are fixed here so that the result can be judged against a standard that was not chosen with the result already in view. Where a threshold is derived from physics, the derivation is given. Where it is a project decision, it is labelled as one and is open to revision, but only by amending this chapter in a dated revision, not silently at analysis time.

2.1 Why this chapter is not optional

A low residual on a degenerate set is not evidence of a good calibration; it is evidence that the ambiguity was absorbed somewhere. The project has already been caught by exactly this: the best archived session yields a root-mean-square reprojection error of 1.50 px, which the existing code classified as merely high rather than catastrophic, while the same fit gives $f_x = 2651.79 \pm 22.42$ px, a 0.85% uncertainty, and a distortion coefficient $k_3 = 3.129 \pm 3.714$ whose standard deviation exceeds its own value [5]. A number that looks acceptable while the parameters behind it are undetermined is the specific failure this chapter is designed to prevent repeating at the level of the whole experiment.

2.2 A1. Calibration

Table 2.1: Acceptance criteria for the calibration session that precedes the experiment. All are evaluated by the guided tool at the end of the session, before any experimental capture is taken.

Id	Criterion	Threshold
A1.1	Accepted views covering each of the four tilt bins (0–10°, 10–30°, 30–45°, 45–70°)	≥ 3 per bin
A1.2	Relative standard deviation of f_x and of f_y	$\leq 0.3\%$
A1.3	Standard deviation of each distortion coefficient, relative to the magnitude of that coefficient	< 1
A1.4	Offset of the principal point from the image centre, as a fraction of image width	$\leq 5\%$
A1.5	Capture resolution of the calibration frames	4056×3040
A1.6	Cameras not moved, and focus and aperture rings not touched, between calibration and the last experimental capture	binary

Derivation of A1.2. The focal length in pixels sets the scale of every reconstructed distance directly: a relative error δ in f_x produces a relative error δ in every depth. At the nominal working distance of 3.5 m, holding the scale error from this source below 1 cm requires $\delta \leq 0.29\%$, which rounds to the registered 0.3%. The archived best case is 0.85%, about three times this.

This is the one threshold in the document that may prove unreachable on this rig. If it is, the correct response is to record the value achieved and state that the scale error is correspondingly larger, not to relax the number afterwards. The figure stands as the requirement the measurement has to be judged against, whether or not it is met.

Derivation of A1.3. A parameter whose standard deviation exceeds its own magnitude is not determined by the data; the fit is free to set it to zero without materially changing the residual. This is a binary sanity check, not a tuned threshold, and the archived fit fails it on k_3.

Derivation of A1.4. A principal point far from the image centre is the signature of the $[t_x, t_y]$ against $[c_x, c_y]$ degeneracy, in which a board translation and a principal-point shift become indistinguishable. The active calibration shows $c_x = 694.8$ on a 1920-pixel-wide image where it should sit near 960, an offset of 265 px or 13.8% of the width. The 5% threshold is a project decision chosen to sit comfortably below the known failure and comfortably above the few-percent decentring a real lens assembly can legitimately show.

On A1.5. The relation $f_x = f/p$ means a calibration is tied to the effective pixel size of the mode it was captured in, and two of the IMX477 modes crop as well as bin, so the result cannot simply be rescaled [7]. Every archived session used 1920×1440, never the full sensor. This criterion exists because the single most common way to waste the session is to calibrate in one mode and capture in another.

2.3 A2. Illumination

Each requirement is taken from the dark-room specification [4], where its derivation from primary sources is given. Reproduced here as pass or fail conditions with the measurement that decides them.

Table 2.2: Illumination acceptance criteria and the measurement that decides each.

Id	Criterion and how it is measured	Threshold
A2.1	Stray light: with the fixture off, at the longest exposure to be used, no scene structure discernible in either camera	nothing visible
A2.2	Single direction: one fixture, or several with parallel axes. No fill, no bounce, no ambient	binary
A2.3	Collimation: width of the penumbra cast by a straight edge held 1 m from a white card at the tether position	≤ 35 mm
A2.4	Uniformity over the working volume, including the depth axis, as $(\max - \min)/(\max + \min)$ on a lux grid	$\leq 5\%$
A2.5	Illuminance at the tether position	≥ 2320 lx
A2.6	Short-term stability over the session	$\leq 2\%$
A2.7	Flicker: white card imaged at the shortest exposure in use, each camera	no banding
A2.8	Colour temperature	5600–6000 K

On A2.3. The real Sun subtends 0.533° and would cast a 9.3 mm penumbra at 1 m [8]. The registered 35 mm corresponds to a source of about 2°, set just outside the 1.9° of the ESA Large Space Simulator [9]. The virtue of this test is that it needs a ruler rather than an optical bench.

On A2.4. This is the class B uniformity of IEC 60904-9, using that standard's own definition [10]. The depth axis is the one that fails first and the one most often omitted from a uniformity check. The procedure in Chapter 5 loops back on failure of this criterion specifically, because moving a fixture to fix uniformity changes the illuminance while dimming it does not change the uniformity.

On A2.5. Derived from the Sony IMX477 datasheet [11] for the fitted 8 mm f/1.6 configuration at the 1°/s reference rate, as the illuminance at which a white surface reaches saturation within the maximum exposure that rate allows. It is a floor, not a target: four stated assumptions in the derivation all push the real requirement upward, and a tether darker than a white card needs more.

On A2.7. A mains-connected fixture modulates at twice the line frequency, giving a 10 ms cycle

at 50 Hz, the same order as the exposures in use, and a rolling shutter turns that into banding [12]. Note that a global-shutter sensor is not immune: it converts the banding into frame-to-frame brightness variation, which is equally destructive for a stereo pair that has to be radiometrically consistent. Tuning the exposure to an exact multiple of the flicker period is not available as a remedy, because the exposure is not free. The fixture itself must not flicker.

2.4 A3. Pipeline

These are evaluated after the session, on the captures, by running the application unmodified.

Table 2.3: Pipeline acceptance criteria. Unless stated otherwise each is evaluated on every stereo pair selected for analysis, and the criterion is met if the stated fraction of pairs passes.

Id	Criterion	Threshold
A3.1	Segmentation yields a tether mask that is a single connected component in both images	$\geq 80\%$ of pairs
A3.2	Endpoint detection returns exactly two endpoints in both images	$\geq 80\%$ of pairs
A3.3	Automatic tracking reaches the far endpoint without operator intervention	$\geq 50\%$ of pairs
A3.4	Fraction of path samples surviving the disparity and depth filters of `compute_wire_metrics`	$\geq 90\%$ per pair
A3.5	Reconstructed arc length against the tape-measured tether length, on pairs where the whole tether is in frame in both cameras	within $\pm 10\%$
A3.6	Straightness, on a pair where the tether was deliberately held taut	≥ 0.97
A3.7	At least one complete run producing an exported point cloud and a full set of metric scalars	≥ 1

On A3.3. The 50% threshold is deliberately lower than the others. Automatic tracking is the stage most exposed to crossings and to breaks in the mask, and the application offers a hand-drawn path as a first-class alternative precisely because the automatic route is not expected to be universally reliable yet. A result below 50% is informative rather than disqualifying: it is the measurement that would justify building the learned branch scorer.

On A3.4. `compute_wire_metrics` discards points with non-positive disparity or with Z outside 0.05 to 50 m *before* computing arc length, so a mid-tether segment that gets filtered makes the reported total length too short with no warning anywhere in the output. The retained fraction therefore has to be recorded alongside the length, or the length cannot be interpreted. This criterion exists to force that.

On A3.5. This is the only criterion in the document that compares against an external measurement. The tether length is measured with a tape before the session and recorded in Chapter 6. The $\pm 10\%$ tolerance is a project decision and is loose on purpose for a first run: it has to accommodate the focal scale error permitted by A1.2, the systematic error of the parametric matching assumption, and the under-reporting caused by A3.4, none of which has ever been measured together on hardware. It is set to detect a gross failure, not to certify an accuracy. Tightening it is the job of the next revision, once there is a measured distribution to tighten it against.

On A3.6. Straightness is reported by the pipeline as the ratio of the endpoint chord to the traced arc length, so a taut tether should return a value just below 1. The 0.97 threshold allows roughly 3% of path wander from sag, mask noise and tracking jitter combined.

2.5 A4. Record

On A4.4. The experiment moves the rig by hand. Moving the focus ring changes the focal length, and f_x is the focal length in pixels, so 1% of drift is 1% of scale error on every reconstructed distance. It is a systematic bias and averaging more frames does not remove it. If the marks have moved by the end of the session, the data is in doubt and A3.5 cannot be interpreted; this is the reason the check is a criterion and not a note.

Table 2.4: Completeness criteria for the session record. These decide whether the experiment can be replicated and rendered, not whether it succeeded.

Id	Criterion
A4.1	Every field of the record sheet in Chapter 6 filled, or explicitly marked not measured with the reason
A4.2	Tether physical pose recorded: approach angle, distance to the camera baseline, and attachment points
A4.3	Fixture make, model, modifier, count and ballast or driver mode recorded
A4.4	Ring marks checked and photographed at the end of the session, and the result recorded
A4.5	Calibration session and all experimental captures committed to the repository, with their per-pair metadata

On A4.5. The Blender validation deliverable records that for its chosen reference pair the capture session and exported point cloud were never pushed to the repository, leaving only screenshots [3]. This criterion exists so that it does not happen twice.

2.6 Overall verdict

The experiment is reported as a **pass** if A1, A2 and A4 are met in full and A3 is met except for A3.3 and A3.5. It is reported as a **qualified pass** if A1 and A4 are met, A2 fails on no more than one of A2.3, A2.4 or A2.8, and A3.7 is met: in that case the captures are usable for geometry but not as the photometric reference for the Blender comparison. It is reported as a **fail** if A1 is not met, because then no quantity in metres means anything, or if A3.7 is not met, because then the pipeline did not complete even once.

Chapter 3

Facility and instrument

3.1 The venue

TV2 is a television studio of the PUCP Faculty of Communications, borrowed for the session through internal services. The project does not own a space that can be made dark at this scale, and the studio offers three things the experiment needs and the laboratory enclosure does not: a room that can be blacked out completely, professional lighting fixtures with the rigging to position and aim them, and enough floor depth to place a fixture far from the subject, which is what the collimation requirement A2.3 ultimately demands.

A studio also brings two hazards that the acceptance criteria are written to catch. Studio lighting is designed for an aesthetic result, so fill light, bounce and multiple sources are the norm and have to be deliberately suppressed, which is criterion A2.2. And studio fixtures are frequently dimmed, often by a method whose modulation frequency is unknown, which is criterion A2.7.

3.2 The instrument, as built

Table 3.1: The camera payload as assembled for this experiment. The integrated support was completed and verified working on 13 October 2026; it is the geometry the calibration must be performed on.

Item	Configuration
Compute	Raspberry Pi Compute Module 5, Cortex-A76 at 2.4 GHz, 16 GB
Sensors	Two Arducam HQ 477 modules, Sony IMX477, 12.3 MP [11]
Lenses	Arducam C1508ZM04, 8 mm, f/1.6 to f/16, C-mount [13]
Baseline	100 mm, fixed by the integrated support
Capture	4056×3040, 12-bit sensor mode, explicit `--mode` argument
Preview	1920×1440, used for guidance only, never for measurement
Frame rate	15 frame/s in the current software configuration
Synchronisation	Sequential software triggering, with the inter-camera delay recorded per pair
Calibration target	ChArUco 5×7, 37 mm squares, 29.6 mm markers, `DICT_6X6_250` [14]

On the calibration target. It is a ChArUco pattern printed on paper and mounted on a rigid folder. Two properties of a target matter and neither is usually verified: flatness and detectability. Detectability is the reason for ChArUco rather than a plain chessboard, since a self-identifying target labels every corner individually and tolerates partial views, which is what makes interactive guidance possible [15]. Flatness has never been measured for this board, and a paper sheet on a folder is the weaker of the two properties. It is recorded here as a known limitation rather than as a controlled parameter.

On synchronisation. Hardware triggering is not exposed through the standard development configuration, so the two cameras are triggered sequentially in software. The camera system report

asserts this is adequate for target motion below roughly 5 cm/s [2]; that bound is an assumption rather than a measured synchronisation error. Hand motion will exceed it. The inter-camera delay is recorded per pair, and pairs whose delay is large are candidates for exclusion from A3.5. For reference, the February 2026 reference pair records a 10.4 ms delay.

3.3 The subject

A physical tether proxy, nominally 1 mm in diameter, strung across the LINKU structure. The structure itself is built by another member of the team and is outside the scope of this document; what matters here is that the tether is attached to it at two points whose positions are recorded, and that the camera support is mounted on it.

The 1 mm figure is the physical proxy actually used in the laboratory. It should not be confused with the 0.8 mm figure that appears in the camera system report, which is explicitly described there as a conservative assumption adopted for the analytical detectability estimate and is not a measurement of anything [2].

3.4 Working distance and what the optics do at it

The nominal working distance is 3.5 m, taken from the Blender replica of the structure. At that distance, with the fitted 8 mm lens on the IMX477, the tool [16] gives the figures in Table 3.2.

Table 3.2: Optical behaviour of the fitted configuration at the nominal 3.5 m working distance, computed for the IMX477 with a 1.55 µm pixel and a one-pixel circle of confusion. These figures do not transfer to the IMX708 B0308 flight sensor, which has a different pixel pitch and a different focal length.

Quantity	**Value**	**Note**
Projected width of a 1 mm tether at 3.5 m	1.47 px	marginally resolved
Depth of field focused at 3.5 m, f/1.6	3.08 to 4.05 m	0.96 m total
Hyperfocal distance at f/1.6	25.8 m	
Depth of field focused at 21 m	11.58 to 112.6 m	tether at 3.5 m is blurred
Airy disc diameter at f/1.6	2.15 µm	1.39 px
Maximum exposure at the 1°/s reference rate	11.1 ms	1 px blur budget
Minimum illuminance to saturate a white card in that exposure	2320 lx	criterion A2.5

Two consequences drive the procedure in Chapter 5.

The depth of field is under a metre. Focusing for a far subject does not merely reduce sharpness at 3.5 m, it moves the near limit of the sharp zone out to 11.6 m and leaves the tether entirely outside it. The focus has to be set by looking at a live image at full resolution, at the real distance, and then locked.

The focus ring carries no usable scale. Its full travel moves the lens by 0.45 mm, and the last 0.003 mm of that, 0.7% of the rotation, covers everything from 21 m to infinity. Setting focus by reading a number off the barrel is not a viable method on this lens.

Chapter 4

Parameters to set and lock

Chapter 2 says what will be judged. This chapter says what has to be set, in what order, and why each setting is what it is. The numeric thresholds are not repeated here; they are in Tables 2.1 to 2.4.

Each entry is marked **Physics** if the Sun or the sensor fixes it and the project cannot negotiate, or **Decision** if it is a project choice that could be revised with an argument.

4.1 Illumination

L1. One direction of light. Physics

In orbit the tether is lit by a single distant source and there is no atmosphere to fill the shadows: the lit side is at full sun and the shadowed side is nearly black. Several fixtures are permitted only if their axes are parallel. Converging beams are forbidden, and this is the most important prohibition in the chapter: two converging sources put two shadows and two specular highlight lines on a target that is one or two pixels wide, which presents the matching stage with a doubled thread, a failure mode it is not designed to handle. Also forbidden are mixing fixture types or colour temperatures along a row, and placing a fixture at one end shining along the tether, which combines grazing incidence with inverse-square falloff along its length.

L2. Daylight colour. Decision

Every space vision laboratory in the reviewed set that declares a colour temperature falls in the daylight range, and the DLR EPOS sun simulator is a 6000 K HMI with CRI 90 [17, 18]. The Sun's effective temperature is 5772 K [8]. This rules out 3200 K tungsten without further discussion. TLCI is requested as a datum because CRI was not designed for cameras [19], but no TLCI threshold is set, because no reviewed source gives one.

L3. Parallel beam. Decision, anchored to physics

Verified with the penumbra test of Section 4.2. There is a second, independent reason for collimation beyond matching the Sun's shadow behaviour: without it, the inverse-square falloff makes the uniformity requirement L4 unachievable at any distance that fits in the room. A point source would need a throw of roughly twenty times the object depth to reach class B uniformity, which is about 10 m for a 0.5 m deep structure and 20 m for a 1 m one. Few rooms offer that.

L4. Uniformity across the working volume, depth included. Decision, from IEC class B

Measured on a lux grid that spans width, height and depth. The depth axis fails first and is the one most often left out.

L5. Enough light at the tether. Physics, from the datasheet

Derived in the dark-room deliverable from the IMX477 datasheet. It is a floor. Four assumptions enter the derivation, all of which push the true requirement upward: the datasheet's 3200 K test spectrum differs from the 5600 to 6000 K in use; the treatment of sensor gain; an on-axis transmittance of unity, ignoring vignetting; and a white card of reflectance 0.9 filling many pixels, where the real tether is darker and narrower.

L6. No flicker. Physics

For an HMI, use an electronic ballast in its high-frequency mode, near 1000 Hz. Note specifically that the ARRI EB 12/18 manual states its "50/60 Hz, low noise" mode is *not* flicker free despite the ballast being an electronic one, and that its 75 Hz "flicker free" mode is also unsuitable here because the resulting period exceeds the exposure ceiling set by the rotation rate [20]. Magnetic ballasts are excluded. For an LED fixture, request the dimming method and the PWM frequency from the supplier; no reviewed source gives a safe value, so the fixture is accepted only after the on-site test A2.7.

L7. Black surroundings, fixture out of frame. Decision

Black velvet or molton, no light falling on walls. This is how DLR and Stanford build their testbeds [18, 21], and black velvet was the best background measured with the same Raspberry Pi HQ camera this project uses [22]. A fixture inside the frame produces flare; Stanford report exactly that when the camera points near their sun lamp. IEC 60904-9 separately warns that wall reflections invalidate a measured uniformity classification [10].

4.2 The penumbra test

Hold a straight edge 1 m in front of a white card placed at the tether position, and measure the width of the diffuse band with a ruler. The geometry is

$$w = L \tan \alpha, \tag{4.1}$$

where L is the edge-to-card distance and α the angular size of the source as seen from the edge. The measured width therefore gives the angular size directly. The Sun would give 9.3 mm at $L = 1$ m; the registered limit of 35 mm corresponds to about 2°; an uncollimated lamp typically gives 87 mm or more. The value of this test is that it converts an optical property into a ruler reading.

4.3 Optics

O1. Aperture wide open at f/1.6. Decision, with calculation

Two independent arguments point the same way. Stopping down costs light as the square of the f-number: going to f/2.8 triples the illuminance requirement, from 2320 to about 7100 lx. And diffraction is already significant wide open, with the Airy disc at 2.15 µm or 1.39 px at f/1.6, rising to 2.42 px at f/2.8 and 4.85 px at f/5.6, against a tether that projects to 1.47 px. There is no aperture at which closing down helps.

O2. Focus at the real tether distance, set by eye. Physics

Set while looking at the live image at full resolution, with the tether in place. See Chapter 3 for why the barrel scale cannot be used.

O3. Lock and mark both rings. Critical

Tighten the grub screws and mark the ring positions with a pen or tape. Moving focus changes the focal length, and f_x is the focal length in pixels, so 1% of drift is 1% of scale error on every reconstructed distance, or 10 mm per metre. It is a systematic bias and averaging does not remove

it. This experiment moves the rig by hand, so a loose ring will move. The marks are checked at the end of the session, which is criterion A4.4.

4.4 Calibration

C1. Same resolution and same sensor mode as the experiment. Physics

Capture the calibration frames at 4056×3040 with an explicit `--mode`. Without that argument libcamera may silently pick a mode and then scale or crop, which makes the transfer untrustworthy [7].

C2. Calibrate on the integrated support. Physics

The stereo extrinsics describe the rigid transform between two specific cameras in specific positions. Calibrating on one mount and then moving the cameras to another invalidates them. The support is already built; the calibration session happens on it, in the room, with the structure in place.

C3. Fill all four tilt bins. Physics

This is the correction for the diagnosed cause of the 23 previous failures. With the board nearly parallel to the sensor in every view, a longer focal length at a greater distance produces an identical image and the focal length is simply not recoverable, however many frames are taken [23, 5]. The guided tool asks for a specific pose, draws its outline in both previews, and only accepts a frame that is on that pose, so filling the bins is the tool's job rather than the operator's; what the operator has to do is physically reach the poses it asks for, including the strongly tilted ones.

C4. Do not touch the optics afterwards. Critical

See O3. Calibration and experiment are one continuous optical configuration.

4.5 Camera settings during the checks and the run

Base gain, fixed white balance, manual exposure at the ceiling given in Table 3.2. Automatic exposure and automatic white balance are both disabled: an auto-exposure loop will chase the moving tether and change the radiometry from frame to frame, which breaks both the flicker test and the stereo consistency the matching depends on.

Chapter 5

Session procedure

5.1 Why the order matters

The sequence below is not a checklist that can be reordered for convenience. The geometric settings are fixed before the photometric ones for a specific reason: **moving a fixture to correct uniformity changes the illuminance, while dimming it does not change the uniformity**. Settle position and aim first, then level. Doing it the other way round means measuring the light twice.

The same logic places calibration after the optics are locked and before any experimental capture, and places the stray-light check before everything, because a room that is not actually dark invalidates every photometric number measured in it.

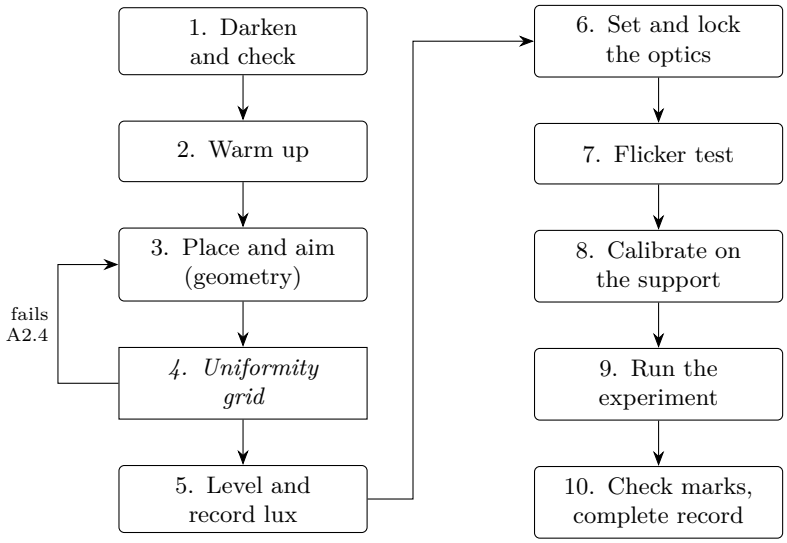


Figure 5.1: The session procedure. The only loop is the uniformity check returning to fixture placement, because correcting uniformity means moving a fixture, which changes the illuminance and invalidates any level already measured.

5.2 The ten steps

1. **Darken the room.** Close, cover, and rig the black background. Verify with a frame from each camera at the longest exposure that will be used, with the fixture off. Nothing recognisable may appear. Anything visible is stray light and must be eliminated before continuing. This is criterion A2.1.

2. **Warm up the fixture** for the manufacturer's stated time before measuring anything. For

HMI this is several minutes. If a spectroradiometer is used, EBU recommends at least 30 minutes for the instrument itself [19].

3. **Place and aim.** Position the fixture at the greatest throw the room allows and set its aim. Run the penumbra test of Section 4.2 and record the width. This is criterion A2.3.
4. **Map the uniformity.** Take a lux grid across width, height and depth of the working volume. If criterion A2.4 fails, move the fixture and return to step 3. Do not attempt to fix uniformity by dimming.
5. **Record the level.** Illuminance at the tether position, against criterion A2.5. Also record the grid minimum and maximum from step 4 and the measured colour temperature.
6. **Set and lock the optics.** Aperture wide open, focus set by eye on the live full-resolution image with the tether in place, then both rings tightened and marked. Photograph the marks.
7. **Flicker test.** White card, shortest exposure in use, one frame from each camera. Look for banding. This is criterion A2.7. If it fails, the fixture or its ballast mode has to change, and steps 3 to 5 are repeated for the replacement.
8. **Calibrate** on the integrated support, in place, following Section 5.3.
9. **Run the experiment** following Section 5.4, touching nothing on the optics.
10. **Close out.** Check the ring marks against the photographs from step 6 and record whether they moved. Complete every field of the record sheet. Copy the captures off the device.

5.3 Running the calibration

The application's guided flow does the bookkeeping. The operator's job is physical.

The tool asks for one pose at a time and draws its outline in both camera previews. A frame is captured only when it is sharp over the board region, momentarily still, visible to *both* cameras, and on the pose currently being asked for. A rejected frame does not consume a slot, so a request for 25 views yields 25 that passed every gate.

Three practical points for the operator:

- **Stand back and hold the board yourself.** The flow is built for a single person holding the board at a distance and matching the drawn outline. The preview is sized so it remains readable from across the room.
- **The strongly tilted poses are the point.** The bins at 30 to 45° and 45 to 70° are the ones no archived session ever reached, and they are the ones that make the focal length recoverable. If a pose cannot be physically reached, use *Finish with the views collected*; that is the deliberate way out, and the session reports itself as incomplete rather than pretending otherwise.
- **Watch the uncertainty, not the residual.** The progress indicator is driven by parameter uncertainty from the fit covariance, not by a frame count and not by the reprojection error. A session is finished when the uncertainty stops falling, not when a number of frames has been reached.

At the end, evaluate Table 2.1. If A1 fails, the experiment can still be run, but Chapter 7 will report lengths in pixels and ratios rather than in metres, and criterion A3.5 is void.

5.4 The capture plan

Both the tether and the rig are moved by hand. The plan is organised so that the simplest configurations are captured first, while everything is known to be working, and the hardest last.

S1 is the measurement set. It is the only one where an external reference exists, because the tether is taut and its length has been tape-measured. Everything else is an exercise of robustness.

Table 5.1: Capture sequence. Each configuration is captured as several stereo pairs, so that a failure in any one pair does not cost the configuration. Counts are a minimum.

Set	Configuration	Pairs
S1	Tether taut and static, whole length in frame in both cameras. The reference set for A3.5 and A3.6	≥ 5
S2	Tether with deliberate sag, static	≥ 3
S3	Tether static, camera rig moved slowly by hand	≥ 5
S4	Tether static, camera rig moved erratically	≥ 5
S5	Tether in motion, rig static	≥ 5
S6	Both in motion, erratic. The nominal case	≥ 10
S7	Tether arranged with a deliberate self-crossing, static	≥ 3
S8	One endpoint deliberately out of frame in one camera only	≥ 3

S7 exists to characterise the tracker, not to pass. A self-crossing is where the automatic tracker has to choose a branch using a hand-tuned scoring formula, and it is the case the project expects to be weakest. Capturing it deliberately turns an anticipated weakness into a measurement, and the pairs feed the path dataset whether the tracker succeeds or not.

S8 exists to provoke a silent failure. The parametric matching assumption breaks when the two traced paths do not cover the same physical extent, and nothing in the pipeline detects it. Deliberately occluding one endpoint in one camera creates that condition on purpose, so that the size and the sign of the resulting error can be seen for the first time against the tape-measured length from S1.

Chapter 6

What gets recorded

The session produces two things: image files, which the software names and stores by itself, and written observations, which nothing records unless a person writes them down. This chapter specifies the second, because it is the part that has previously gone missing.

6.1 The record sheet

Every field is filled during the session, not reconstructed afterwards. A field that could not be measured is marked *not measured* with the reason, which is criterion A4.1. An empty cell is indistinguishable from a forgotten one.

Table 6.1: The session record sheet. Transcribed into the results chapter after the session.

Datum	Unit	Value
Venue and date		
Room, date, start and end time		
People present and their roles		
Fixture		
Make, model, modifier, count	text	
Ballast or driver, and the mode selected	text	
Dimming method and PWM frequency, if LED	text, Hz	
Warm-up time allowed before measuring	min	
Distance and angle of the fixture to the structure and to the cameras	m, °	
Photometry		
Illuminance at the tether position	lx	
Uniformity grid: minimum and maximum, and the grid spacing used	lx, m	
Measured colour temperature	K	
Measured CRI, and TLCI if available		
Penumbra width at 1 m	mm	
Stability over the session: minimum and maximum	lx	
Stray-light check result	pass/fail	
Flicker test result, per camera	pass/fail	
Geometry of the scene		
Tether physical length, tape measured	m	
Tether diameter, measured	mm	
Tether material and surface finish	text	
Attachment points on the structure, described or sketched		

Datum	Unit	Value
Tether approach angle relative to the camera baseline	°	
Distance from the camera baseline to the nearest and farthest tether point	m	
Working volume of the structure: width, height, depth	m	
Optics and cameras		
Focus distance set, and how it was judged	m, text	
Aperture set	f/	
Both rings locked and marked, before the session	yes/no	
Both marks unmoved, after the session	yes/no	
Per camera: exposure, gain, white balance	ms, dB	
Capture resolution and sensor mode used	px	
Calibration		
Session directory name	text	
Accepted views per tilt bin	count × 4	
Relative standard deviation of f_x, f_y	%	
Principal point offset from centre	px, %	
Reported reprojection RMS, for the record only	px	
Session reported complete by the tool	yes/no	
Captures		
Capture sets obtained, against Table 5.1	S1–S8	
Total stereo pairs	count	
Inter-camera delay: median and worst	ms	
Anything that went wrong, in prose		

6.2 Why the scene geometry fields are there

The block headed *Geometry of the scene* is not documentation for its own sake. The Blender validation deliverable selected a real stereo pair as its reference and then found it could not reproduce the scene, because the tether's approach angle, distance and attachment points had never been written down [3]. Those four fields exist so that this session's captures do not become unusable for the same reason.

The photometry block serves the same deliverable from the other side: its laboratory-condition light is currently a placeholder, and these measurements are what replaces it.

6.3 Files, naming and metadata

The application writes each stereo pair to its own timestamped directory under `data/captures/`, with per-pair metadata including the capture resolution, the baseline and the inter-camera delay. That naming is automatic and should not be changed by hand. What has to be added is the mapping from directory to capture set.

- Keep a plain text file in the session folder mapping each capture directory to its set S1 to S8 and to a one-line description of what was being done at the time. Written during the session, not after.
- Commit the calibration session directory and every capture directory, with their metadata, which is criterion A4.5. The repository already versions `data/captures/` and `data/calibration/`, so this is the default behaviour rather than an exception, but it has to actually be done.
- Photograph the physical setup: the fixture position, the structure with the tether strung, the camera support in place, and the ring marks. These go in `Images/` of this deliverable.

6.4 What is deliberately not recorded

The angular velocity of the hand motion. There is no instrumentation for it, and writing down an estimate would create a number that later analysis would be tempted to treat as a measurement. Section 1.5 gives the reasoning. The rolling-shutter deliverable is where that quantity gets measured, with a stage that either commands or encodes the rate [6].

Chapter 7

Analysis plan

The analysis is run after the session, on the committed captures, using the application without modification. Running a modified pipeline on the experimental data would mean the result does not describe the software that exists.

7.1 The pipeline, stage by stage

The measurement path is the *Geometric (Wire Tracking)* mode, which is the application default. The alternative dense stereo mode is not used, and Section 7.5 explains why that matters more than it sounds.

1. **Segmentation.** The cable detection tuner produces a binary mask for each image. Two segmenters are available: the classical one, based on a top-hat transform with adjustable parameters, and a trained random forest operating per pixel on nine hand-crafted features. Record which was used for each pair.
2. **Endpoint detection.** The mask is skeletonised and pruned, and the two endpoints are taken as the geodesically farthest pair.
3. **Path tracing.** Either automatic, a depth-first search over the mask with backtracking and momentum, or hand drawn by the operator clicking along the tether. Record which, per pair: this is what criterion A3.3 counts.
4. **Rectification** of both images from the calibration.
5. **Matching.** The point at $i\%$ along the left path is paired with the point at $i\%$ along the right path.
6. **Triangulation** into a sparse point cloud, written out directly from the disparity and the baseline rather than through OpenCV's triangulation routines.
7. **Metrics.** Arc length, straightness, depth range, diameter and curvature.
8. **Export** to PLY, XYZ, PCD or OBJ.

7.2 What is computed, and against what

7.3 The retained fraction has to be reported with every length

`compute_wire_metrics` discards samples whose disparity is non-positive or whose depth falls outside 0.05 to 50 m, and it does this *before* summing the chord lengths. A tether segment that gets filtered out therefore shortens the reported arc length, and nothing in the output says so. A length quoted without its retained fraction is not interpretable, so the two are always reported together. Criterion A3.4 sets the floor at 90%.

Table 7.1: What is extracted from each analysed pair and what it is compared against.

Quantity	Source	Compared against
Mask connectivity	stage 1	criterion A3.1
Endpoint count	stage 2	criterion A3.2
Tracking mode used, and coverage reached	stage 3	criterion A3.3
Retained sample fraction	stage 7	criterion A3.4
Arc length in metres	stage 7	tape measure, criterion A3.5
Straightness	stage 7	criterion A3.6, on taut pairs only
Depth range	stage 7	scene geometry record
Reported diameter	stage 7	measured tether diameter
Curvature	stage 7	qualitative, sag pairs

7.4 The comparison that is expected to fail, and why that is useful

Capture set S8 deliberately places one endpoint outside the frame of one camera. The matching stage assumes both traced paths span the same physical extent of the tether, and there is no epipolar check anywhere in the pipeline that could detect a violation. The predicted consequence is a systematic error in the reconstructed length with no accompanying warning.

The analysis of S8 is therefore: take the arc length reported for the S8 pairs, compare it against the tape-measured length and against the S1 pairs of the same tether, and record the magnitude and the sign of the discrepancy. This is the first time the project would have a measured size for an error it has only ever reasoned about.

A secondary analysis, if the S8 result is large, is to run the same pairs through the epipolar matcher that exists in the codebase but is not reachable from the interface. That matcher does not rely on the parametric assumption and was kept specifically against this possibility. It has never been validated on real data, so a result from it is exploratory, not a measurement.

7.5 Why dense stereo is not used, and the trap that creates

Dense stereo fails on a thin, textureless object because there is nothing along the wire to correlate. This has already been demonstrated twice: commercial structure-from-motion on the isolated tether returns a zero resolution error and produces no camera alignment or sparse cloud at all, the stated reason being insufficient visual information within the tether itself rather than background interference; and dense stereo inside the laboratory enclosure gives correct disparity on a solid target while also producing non-physical disparity throughout regions that should be empty background, yielding phantom geometry that needs manual intervention to clean [2].

The trap is operational. If the operator does not configure a mask in the segmentation tuner, the processing falls back to dense stereo *automatically and silently*. It does not raise an error. It produces something that looks like a point cloud. Any analysis run after skipping the tuner is therefore running the method already known to fail, in a form that looks plausible. The analysis log records, for every pair, that a mask was present.

7.6 Feeding the datasets

Every pair that is successfully traced, automatically or by hand, is saved into the path dataset together with its mask and its verified path. This dataset currently holds three stereo samples against a recommended 30 to 100, and it is the input to the one model in the plan that does not yet exist: a learned per-branch scorer that would replace the fixed hand-tuned weights in the tracker's decision at each crossing.

One caution applies to the other dataset. Saving a sample to the mask dataset while the random-forest segmenter is active means training the model on its own unverified output, which reinforces its errors. The application asks for confirmation in that case; the answer during this session is no, unless the mask has been inspected and corrected.

Chapter 8

Results

This chapter is empty. No result exists yet.

At the time of this revision the repository contains no capture from the TV2 session. The most recent non-simulated stereo pairs in `data/captures/` date from February 2026, and the most recent simulated ones from September 2026. The record sheet of the pre-session parameter artifact has not been filled in.

Nothing in this chapter should be written before the session. What follows is the structure to fill, so that the writing afterwards is transcription rather than reconstruction.

8.1 Session record

Transcribe Table 6.1 here, completed. Every *not measured* entry keeps its reason.

8.2 Calibration outcome

Report against Table 2.1: the accepted view count per tilt bin, the relative standard deviations of f_x and f_y, the standard deviation of each distortion coefficient against its own magnitude, the principal point offset, and whether the tool reported the session complete.

Report the reprojection RMS as well, but report it *after* the uncertainties and label it as the training residual it is. If A1.2 was not met, state the value achieved and carry the implied scale error forward into every length quoted in this chapter.

8.3 Illumination outcome

Report against Table 2.2, one row per criterion, with the measured value and pass or fail. Where a criterion failed, state what was tried.

8.4 Pipeline outcome

Report against Table 2.3, aggregated across the analysed pairs, with the per-set breakdown from Table 5.1. Every arc length is reported with its retained sample fraction beside it.

8.5 The two anticipated failure modes

Parametric matching, from set S8. The measured discrepancy against the tape-measured length, with its sign, and how it compares to the S1 pairs of the same tether.

Rolling-shutter shear, from sets S4 and S6. A qualitative description of whether shear was visible at all, and at what point in the motion the tracker or the matching visibly stopped working. No angular rate is quoted, because none was measured.

8.6 Verdict

State the overall verdict using the definitions at the end of Chapter 2: pass, qualified pass, or fail. State it before discussing what it means.

8.7 What the next session should change

Written last, from what actually went wrong rather than from what was anticipated here.

Pending: run the session and complete this chapter. Until it is complete, this deliverable is a protocol, not a report, and should not be cited as evidence that the instrument has been validated.

Status and limitations

Where each part stands

Table 8.1: What is settled and what is not.

Part	Status	Basis
Objectives	Settled	Derived from the work plan package and from the gaps the companion deliverables leave open
Acceptance criteria	Registered	Fixed before any data exists. A1.2, A2.x and A3.4 derived; A1.4, A3.3, A3.5 and A3.6 are project decisions
Instrument description	Measured	The hardware exists and was verified working on 13 October 2026
Optical figures	Derived	Thin-lens model for the IMX477 with the fitted 8 mm lens; reproduced independently from first principles
Illumination requirements	Derived	From primary sources in the companion dark-room deliverable
Procedure	Settled	Ordering follows from the physics, not from convenience
Record sheet	Settled	Scope set by what the Blender comparison needs
Analysis plan	Settled	Runs the existing application unmodified
Results	**Absent**	The session has not been recorded

Limitations that are known before the session

No angular rate is measured. Both the tether and the rig are moved by hand. Nothing in this experiment can produce a quantitative rolling-shutter result, and the document is written so that nothing in it can be mistaken for one.

No independent ground truth except length. The only external measurement available is a tape-measured tether length, which constrains scale but says nothing about shape. Straightness and curvature are judged for internal consistency and plausibility only.

Board flatness is unverified. The calibration target is paper on a folder. Flatness is one of the two target properties that matter and it has never been measured for this board.

The sensor is not the flight sensor. Every optical number here is computed for the IMX477 with its 1.55 µm pixel and the 8 mm lens. The decided flight sensor is the IMX708 B0308, which has a different pixel pitch and a different focal length, and which has not been purchased. None of these figures transfers to it without recomputation.

Synchronisation is sequential and software-timed. The adequacy bound of roughly 5 cm/s quoted in the camera system report is an assumption, not a measured synchronisation error, and hand motion exceeds it.

What this experiment unblocks

If it passes, three things that are currently waiting can proceed. The Blender validation gets measured laboratory light values and a recorded tether pose, which are its two missing inputs. The path dataset gains the first real samples of crossings and of partial occlusion, which is the input to the learned branch scorer. And the project gains, for the first time, a reconstruction length with a stated uncertainty that can be compared against something measured with a tape.

If it fails, the most likely place is A1, the calibration, which is the project's longest-standing open problem and the one with the least margin in the schedule.

Appendix A

Constants and their provenance

Every number used in this document, with where it came from. The status column distinguishes a measurement from a calculation from an assumption, because the three carry different weight and the distinction is lost once they share a table in a slide.

Table A.1: Constants and derived quantities used in this document.

Quantity	**Value**	**Status and source**
The Sun		
Total solar irradiance, AM0	1361 W/m^2	IAU 2015 nominal [8]
Solar illuminance in orbit	132 800 lx	Calculated, AM0 spectrum against $V(\lambda)$ [24, 25]
Solar effective temperature	5772 K	IAU 2015 nominal [8]
Solar angular diameter	0.533°	IAU radius and distance [8]
Penumbra cast by the Sun	9.3 mm per m	Calculated from the above, Eq. 4.1
Sensor and lens, as fitted		
Sensor	Sony IMX477	Hardware [11]
Pixel pitch	1.55 µm	Datasheet [11]
Full sensor resolution	4056 × 3040	Datasheet
Focal length	8 mm	Lens [13]
Aperture range	f/1.6 to f/16	Lens [13]
Horizontal field of view	42.9°	Calculated from active area and focal length. **Not** the 58.4° on the lens datasheet, which is for a 2/3” format
Stereo baseline	100 mm	Integrated support
Fitted f_x, best archived session	2651.79 ± 22.42 px	Measured, at 1920 × 1440 [5]
Scene		
Nominal working distance	3.5 m	From the Blender replica, not a measured room dimension
Tether diameter, physical proxy	1.0 mm	Laboratory item
Tether diameter, analytical worst case	0.8 mm	**Assumption** stated as such in [2]. Do not confuse with the proxy
Tether length, mission design figure	up to 100 m	Mission design, not measured
Derived optics at 3.5 m		
Projected tether width	1.47 px	Calculated [16]
Depth of field at f/1.6	3.08 to 4.05 m	Calculated, 1 px circle of confusion [16]
Hyperfocal distance at f/1.6	25.8 m	Calculated
Airy disc at f/1.6	2.15 µm, 1.39 px	Calculated
Motion and exposure		

Quantity	Value	Status and source
Reference angular rate	1.0°/s	Corresponds to the software's current 15 frame/s configuration, **not** a sensor limit [6]
Sensor-limited angular rate	2.76°/s	12-bit full-resolution readout. Itself a **bound**: the exact row-scan time needs the IMX477 Software Reference Manual, which is not public
Shear budget	3 px	**Assumption**, chosen because the tether is 1 to 3 px wide
Blur budget, this document	1 px	Project choice, deliberately stricter than the 3 px above
Maximum exposure at 1°/s	11.1 ms	Calculated
Illumination requirements		
Minimum illuminance, fitted configuration	2320 lx	Derived from the datasheet; a **floor** [11, 4]
Minimum illuminance, Hawk-eye	3620 lx	**Assumption**: the IMX477 saturation rescaled by aperture, because Arducam does not identify the sensor die
Penumbra limit	35 mm at 1 m	Project decision, near the ESA 1.9° reference [9]
Uniformity limit	5%	IEC 60904-9 class B [10]
Stability limit	2%	IEC 60904-9 class B short term
Colour temperature	5600 to 6000 K	Project decision, from laboratory practice [17]
Calibration target		
Board	ChArUco 5×7	`DICT_6X6_250` [14]
Square and marker size	37 mm, 29.6 mm	Generated by the repository tool
Tilt bins	0–10, 10–30, 30–45, 45–70°	Project split, following the ±70° range of [15]
Minimum views per bin	3	Project decision

Appendix B

Sources, with their provenance level

Each cited source is archived in the `Sources/` folder of this deliverable. Where a source could not be obtained in full, or is secondary, that is stated here rather than buried in a footnote, because a reader checking a number is entitled to know how firm the ground under it is.

Table B.1: Provenance of the cited sources.

Source	Level	Note
IEC 60904-9:2020 [10]	**Partial**	Only the free official preview was read, clauses 1 to 5.2, which does include all three class tables. The measurement procedures after 5.2 remain behind the paywall. Every class threshold adopted rests on this partial read
ASTM E490 AM0 spectrum [24]	Primary, mirrored	The data file is taken from a public mirror; the original NREL path is dead
CIE $V(\lambda)$ [25]	Primary	
IAU 2015 nominal solar values [8]	Primary	
Sony IMX477 datasheet [11]	Primary	Sections 10-1 and 11-3/11-4 give the sensitivity and saturation figures used. The Software Reference Manual, which holds the exact readout timing, is not public
Arducam C1508ZM04 [13]	Vendor	Its quoted 58.4° horizontal field of view applies to a 2/3" format and must not be used with this sensor
Raspberry Pi camera documentation [7]	Primary	Source of the sensor mode behaviour, including that some modes crop as well as bin
ARRI EB 12/18 manual [20]	Primary	The statement that the 50/60 Hz mode is not flicker free is the manufacturer's own
EBU Tech 3355, TLCI [19]	Primary	
Sheinin et al., rolling shutter on the grid [12]	Primary	
ESA Large Space Simulator [9]	Primary, web	Beam specifications from the agency's own page
DLR EPOS portfolio [17]	Primary	Operator document. Note that a widely cited survey gives the fixture designation incorrectly as ARRI Max 18/12; the operator documents say 12/18
Benninghoff et al., EPOS 2.0 [18]	Primary	

Source	Level	Note
Park et al., TRON [21]	Primary	Operator paper. A web summary describes the lamp as a xenon short arc; the operators state a metal halide arc
Pauly et al., lessons from a space lab [22]	Preprint	The arXiv v2 text is the version read, which differs from the later journal version
Zhang, flexible camera calibration [23]	Primary	
Rojtberg and Kuijper, pose selection [15]	Primary	Source of the tilt range the bins cover
Garrido-Jurado et al., ArUco [14]	Primary	
OpenCV calib3d [26]	Primary	
LINKU companion deliverables [2, 4, 5, 6, 3, 1]	Internal	Two of these are working drafts whose results chapters are pending; the camera system report contains no reconstruction accuracy figure
`camera_dof_visualizer.py` [16]	Internal tool	Thin-lens model. The depth-of-field figures quoted in this document were reproduced independently from the thin-lens equations as a check

Bibliography

[1] LINKU Project, "LINKU camera payload master work plan," tech. rep., Pontificia Universidad Catolica del Peru (PUCP) / APSCO. Internal project deliverable, `Documentation/LINKU_Deliverable_Master_WorkPlan`. Defines this experiment as the package *Experimento de caja negra, hilo en movimiento* inside the phase-2 deliverable *Integracion y experimento en la estructura LINKU*.

[2] LINKU Project, "LINKU project subdeliverable d5: Camera system report, version 2," tech. rep., Pontificia Universidad Catolica del Peru (PUCP) / APSCO. Internal project deliverable, `Documentation/LINKU_Project_SubDeliverable_D5_CameraSystem_Report__v2_`. Chapter 5 records the 8 mm Arducam C1508ZM04 as the lens fitted to the experimental stereo pair.

[3] LINKU Project, "Blender simulation validation for the LINKU camera system," tech. rep., Pontificia Universidad Catolica del Peru (PUCP) / APSCO. Internal project deliverable, `Documentation/LINKU_Deliverable_Blender_Simulation_Validation`. Consumes the illumination and pose values recorded by this experiment as its laboratory-condition scene parameters.

[4] J. Sifuentes, "Dark-room solar illumination specification for tether imaging," tech. rep., Pontificia Universidad Catolica del Peru (PUCP) / APSCO, LINKU Project, 2026. Companion deliverable, `Documentation/LINKU_Deliverable_DarkRoom_Illumination`.

[5] LINKU Project, "A guided, self-evaluating stereo calibration procedure for the LINKU camera pair," tech. rep., Pontificia Universidad Catolica del Peru (PUCP) / APSCO. Internal project deliverable, `Documentation/LINKU_Deliverable_Calibration_Method`. Diagnoses the archived calibration failures as pose degeneracy and specifies the guided replacement implemented in `camera/`.

[6] LINKU Project, "Rotation and rolling-shutter experiment for the LINKU camera system," tech. rep., Pontificia Universidad Catolica del Peru (PUCP) / APSCO. Internal project deliverable, `Documentation/LINKU_Deliverable_Rotation_RollingShutter_Experiment`. Source of the angular-rate limits used here. Working draft: its chapters 3 and 4 are pending execution.

[7] Raspberry Pi Ltd, "Raspberry pi documentation: Camera (hardware specification, global shutter camera, filters)." https://www.raspberrypi.com/documentation/accessories/camera.html; source files from https://github.com/raspberrypi/documentation. Accessed October 2026; source files saved in Sources/raspberrypi_camera_docs/.

[8] A. Prša, P. Harmanec, G. Torres, *et al.*, "Nominal values for selected solar and planetary quantities: IAU 2015 resolution B3." The Astronomical Journal 152:41 (2016). arXiv:1605.09788. https://arxiv.org/abs/1605.09788, 2016.

[9] European Space Agency, "Large space simulator." ESA Technology, ESTEC Test Centre. https://technology.esa.int/page/large-space-simulator. Accessed October 2026; page saved in Sources/.

[10] International Electrotechnical Commission, "IEC 60904-9:2020 photovoltaic devices - part 9: Classification of solar simulator characteristics, edition 3.0." Official preview distributed via iTeh Standards. https://cdn.standards.iteh.ai/samples/22246/3f3a2edfed5a48678baaf60bebfdd273/IEC-60904-9-2020.pdf, 2020.

[11] Sony Semiconductor Solutions Corporation, "IMX477-AACK-C: Diagonal 7.857 mm (type 1/2.3) 12.3 mega-pixel cmos image sensor with square pixel for color cameras," Tech. Rep. E17203, Sony Semiconductor Solutions Corporation. Sections 10-1 (image sensor characteristics) and 11-3/11-4 (standard imaging conditions, measurement methods). Hosted at https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/source/IMX477-DS.pdf.

[12] M. Sheinin, Y. Y. Schechner, and K. N. Kutulakos, "Rolling shutter imaging on the electric grid," in *IEEE International Conference on Computational Photography (ICCP)*, 2018. https://www.cs.toronto.edu/~kyros/pubs/18.iccp.rolling-ac.pdf.

[13] Arducam, "C1508ZM04 / LN043: C-mount lens for raspberry pi high quality camera, 8 mm focal length, manual focus, adjustable aperture." Product specification. https://www.uctronics.com/arducam-44-degree-2-3-c-mount-8mm-for-raspberry-pi-high-quality-camera.html. Optical format 2/3"; F1.6–F16; MOD 0.15 m; back focal length 17.53 mm; FOV 58.4°(H)×44.6°(V) on 2/3", 44°(H) on the 1/2.3" IMX477. Accessed October 2026; page saved in Sources/.

[14] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez, "Automatic generation and detection of highly reliable fiducial markers under occlusion," *Pattern Recognition*, vol. 47, no. 6, pp. 2280–2292, 2014. The ArUco system underlying the ChArUco board. Copy in Sources/.

[15] P. Rojtberg and A. Kuijper, "Efficient pose selection for interactive camera calibration," in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 31–36, 2018. Fraunhofer IGD / TU Darmstadt. Open-source implementation: https://github.com/paroj/pose_calib. Preprint arXiv:1907.04096, copy in Sources/.

[16] LINKU Project, "camera_dof_visualizer.py: optical comparator for the LINKU candidate sensors." Repository tool, `calculate_cam/camera_dof_visualizer.py`, also embedded as the Optics page of the application. Thin-lens depth of field, hyperfocal distance, diffraction spot and projected thread width, evaluated per sensor and lens. Source of the depth-of-field figures quoted in this document.

[17] German Aerospace Center (DLR), "Space operations and astronaut training portfolio: EPOS, document 2025-06-27_rb_sp_epos_v01." https://www.dlr.de/en/rb/research-operation/portfolio/dlr_rb_portfolio_epos.pdf/@@download/file, 2025. Section 2.2, Sun Simulator, including Fig. 2-4.

[18] H. Benninghoff, F. Rems, E.-A. Risse, and C. Mietner, "European proximity operations simulator 2.0 (EPOS) - a robotic-based rendezvous and docking simulator," *Journal of Large-Scale Research Facilities*, vol. 3, p. A107, 2017. DLR. Identifier printed on the paper: jlsrf-3-155-6. Open access copy: https://elib.dlr.de/111852/1/Final_Version_Online.pdf.

[19] European Broadcasting Union, "Method for the assessment of the colorimetric properties of luminaires: The television lighting consistency index (TLCI-2012) and the television luminaire matching factor (TLMF-2013)," Tech. Rep. Tech 3355, EBU, Geneva, 2017. https://tech.ebu.ch/docs/tech/tech3355.pdf.

[20] ARRI, "Operating instructions, electronic ballast EB 12/18 kw high speed, L2.76296KH." https://www.arri.com/resource/blob/31462/53969eea82ce888fcdcdd510e9f9eca3/l5-0013084-arri-eb-12-18-operating-instructions-en-data.pdf. Accessed October 2026.

[21] T. H. Park, J. Bosse, and S. D'Amico, "Robotic testbed for rendezvous and optical navigation: Multi-source calibration and machine learning use cases," in *AAS/AIAA Astrodynamics Specialist Conference*, no. AAS 21-654, 2021. Stanford Space Rendezvous Laboratory. https://slab.sites.stanford.edu/sites/g/files/sbiybj25201/files/media/file/asc2021_parkbossedamico_final.pdf.

[22] L. Pauly, M. L. Jamrozik, M. Ortiz del Castillo, O. Borgue, I. P. Singh, M. R. Makhdoomi, O.-O. Christidi-Loumpasefski, V. Gaudillière, C. Martinez, A. Rathinam, A. Hein, M. Olivares-Mendez, and D. Aouada, "Lessons from a space lab – an image acquisition perspective." arXiv preprint arXiv:2208.08865v2, SnT, University of Luxembourg. Published 2023 in International

Journal of Aerospace Engineering, doi 10.1155/2023/9944614; the arXiv v2 text is the version read. https://arxiv.org/abs/2208.08865, 2022.

[23] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000. Also Microsoft Research Technical Report MSR-TR-98-71. Copy in Sources/.

[24] ASTM International, "ASTM e490-00a standard extraterrestrial (AM0) solar spectral irradiance, tabulated data." Data file e490_00a.dat, mirrored in https://github.com/pytroll/pyspectral; original at http://rredc.nrel.gov/solar/spectra/am0/, 2000.

[25] International Commission on Illumination (CIE), "CIE spectral luminous efficiency for photopic vision, V(λ)." CIE 018:2019, Table 1. Dataset DOI 10.25039/CIE.DS.dktna2s3. https://cie.co.at/datatable/cie-spectral-luminous-efficiency-photopic-vision, 2019.

[26] OpenCV, "Camera calibration and 3d reconstruction, OpenCV documentation." https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html. Reference for `calibrateCamera`, `stereoCalibrate`, `calibrateCameraRO` and the ChArUco functions used by this project. Accessed October 2026.